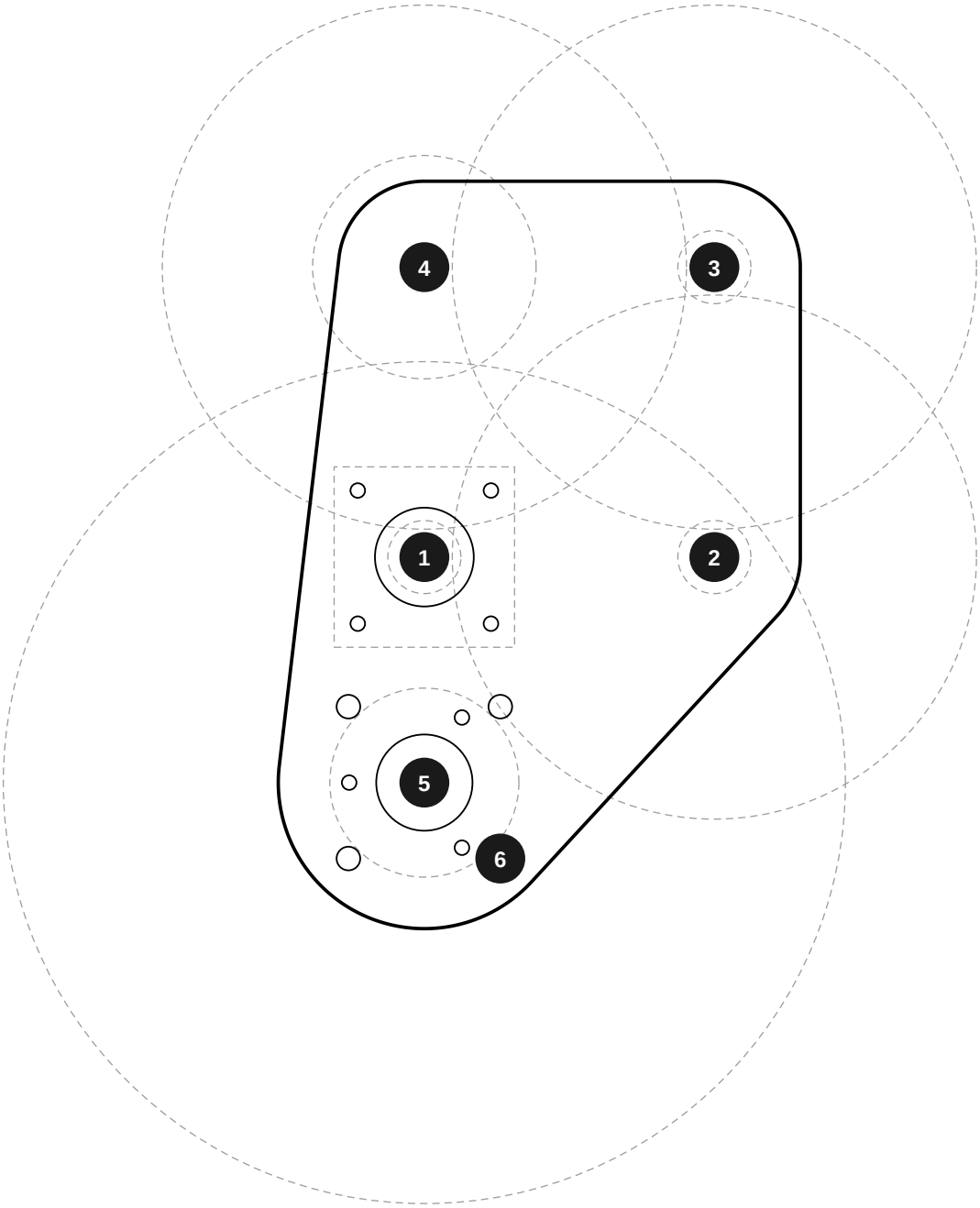
Baseplate assembly guide

NOT TO SCALE - drill from baseplate_template.pdf



Dashed = parts that sit above the plate. They overhang it by design.

#	Shaft	Spacer	Sits on it (bottom to top)	Fasteners
1	S0 motor	-	NEMA17 below plate; pinion.stl above	4 x M3x8 flat head, countersunk
2	S1	2 mm	spacer, then stage.stl (120T + 15T)	20d nail shaft, head under the plate
3	S2	8.5 mm	spacer, then stage.stl (120T + 15T)	20d nail shaft, head under the plate
4	S3	15 mm	spacer, then final_stage_gear.stl	20d nail shaft, head under the plate
5	S4 output	-	tower halves clamp the plate, 2x625ZZ, shaft, washer, hubs, 96T wheel	3 x M3x14 into the disc; 6 x M3x20 hub clamp
6	wedge	-	tripod / wedge adapter, below plate	4 x M5 on dia 50

Notes

Only THREE of the five printed spacers get used - S1 2.0, S2 8.5, S3 15.0. S0's would fall through the dia 23 motor bore (pinion height is set by its grub screw); S4's job is done by the bearing tower.

Nails go in from UNDERNEATH, head against the plate's underside. Cut the point off and square the end.

Trim the NEMA17 shaft to ~10 mm above the plate. At full 24 mm length it fouls the 96T output wheel.

Countersink the 4 motor holes: socket-cap heads hit the S1 wheel 2.0 mm above the plate. M3x8 flat head leaves 3.7 mm of thread; M3x10 bottoms out in the motor.

The M5 wedge hole nearest S1 is also under the S1 wheel: countersink it or leave that bolt out.

Nothing over dia 13 mm may sit inline at S1-S3 - a stock 5 mm shaft collar will hit the neighbouring wheel.

Rewind then approach the start angle FORWARDS; the train has ~750 arcsec of backlash at the output.